Sanchit Rathore

B.Tech – Instrumentation and Control Engineering
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Educational Qualification

Year	Degree/Examination	Institution/Board	CGPA/Percentage
2023-Present	B.Tech- ICE	NIT, Trichy	8.98
2022	Class XII	Mahar Regiment Public School, Sagar, Madhya Pradesh, CBSE	93.80%
2020	Class X	Mahar Regiment Public School, Sagar Madhya Pradesh, CBSE	94.16%

Academic Achievements

- Achieved Pupil rank on Codeforces.
- Solved 1,000+ problems on LeetCode with a rating of 1788.

Internship Experience

SAP Labs, SDE Intern – Joule *May 2026 – July 2026*

- Built an **AI-native multi-agent** application that automated the production incident remediation pipeline reducing analysis time using Python, **Temporal workflows**, **LLM integration**, and **prompt engineering**.
- Designed a **fault-tolerant** remediation pipeline that ingested distributed trace logs from **Kibana**, performed semantic retrieval over **historical incident memory**, and used prompt engineering with Claude to generate root-cause analysis and remediation suggestions for **distributed microservices**.
- Deployed on **Kubernetes**, isolating untrusted LLM-generated fixes in a pod powered by **gVisor-based sandbox** for automated unit testing, with a read-only **HANA database** access via an **MCP Server**.
- Implemented multi-agent deep triage with **Amazon S3**-backed learning memory using **Open Knowledge Format (OKF)** by providing the agent with a Bash tool for executing **grep/ls/cat** commands over the memory directory for improving accuracy, plus a **FastAPI and Redis**-based real-time dashboard with **one-click fix** and **automated pull requests** to GitHub Enterprise.

Projects

- **CodeStein** *March 2026 – April 2026*
  - Developed CodeStein, an online code judge platform where users submit code and receive execution-based verdicts, including challenges such as **Blind Coding and Least Character Count.**
  - Engineered an asynchronous evaluation pipeline using **Go backend, PostgreSQL, and RabbitMQ**, where submissions are stored, queued, and securely dispatched to isolated code runners.
  - Built language execution workers for **Python and C**, processing queued submissions, returning results tagged with submission IDs, while server APIs used polling to fetch verdicts from database.
  - Implemented secure sandboxed code execution with **nsjail**, handling **MLE and TLE** and automated judging against predefined database test cases to generate **Accepted / Wrong Answer** verdicts.

- **Intern Book** *Nov 2025 – Jan 2026*
  - Built a cloud-based web application for NIT Trichy students, architecting a normalized **MySQL** schema and a **Spring Boot backend** exposing **RESTful HTTP APIs**. Integrated **NIT Trichy's OAuth provider** (dAuth) for institution-verified access control.
  - Built an API integration pipeline using **OpenAPI Generator** to convert backend endpoints into a YAML specification for auto-generated API functions, and optimized data access using **EntityGraph** and custom **JPQL to reduce N+1** queries.
  - Implemented a **rate limiter** using the **Token Bucket algorithm** to mitigate spam requests, with separate buckets per IP and a global bucket to protect overall server capacity. Containerized the application with **Docker**, configured **Nginx as a reverse proxy**, and deployed on **Microsoft Azure.**
  - Developed an **infinite scroll** feature using **cursor-based pagination**. Used **React Query** for cached API requests and efficient data fetching, added a **global search** capability using a **Trie-based algorithm** to enable fast keyword lookup across submissions.
- **Attack on Robots** *Nov 2024 – Feb 2025*
  - Built a 2D multiplayer resource management robot battle game **(300+ players)** with **React.js, Phaser.io, and TypeScript** for frontend, and **Rust with PostgreSQL** for backend, where players can design bases by plotting roads, buildings, and mines.
  - Implemented a validator system with **WebSockets** that sends game states from the frontend every tick, validated by the Rust server to maintain accuracy. Utilized **graph algorithms** to compute shortest path between defenders and player for defenses
  - Developed a real-time mini-map showing live player movements, roads, and buildings, with a UAV mode to temporarily reveal hidden defenses like mines. Designed an **AI system using Gemini**, generating live taunts from parsed game events.

Technical Skills

- Programming Languages : C++, Python, Java, JavaScript (JS), TypeScript (TS)
- Web Technologies : HTML, Canvas, CSS, Tailwind CSS
- Frameworks and Libraries : Spring Boot, React.js, Next.js, Node.js, Express.js
- Developer Tools : Git, Postman, Docker, Kubernetes
- Database : MongoDB, MySQL, PostgreSQL
- Other : Linux(Basics), OS, DBMS, OOP

Positions of Responsibility

- **Software Developer, Delta Force Club:** *Aug 2024 - Present*
  Worked as a Software Developer at Delta Force (the official Programming and Web Team of NIT Trichy), collaborating with cross-functional teams in an agile environment, participating in **design discussions**, **peer code reviews**, and the **full software development lifecycle (SDLC)** to deliver scalable web solutions.
- **Manager, Pragyan Webops Team:** *Dec 2024 - Present*
  Worked as the Manager of the Webops Team for Pragyan 2025 (The International Techno - Management Festival of NIT Trichy). Engaged in game development, creating a 2D animated online game for the Pragyan gaming competition. Also contributed to the development and maintenance of the main Pragyan website.

Extracurricular Activities

Cultural Activities:

- Secured **Second** position in **Debate** competition at Aaveg'23 – the inter-hostel cultural fest at NITT.

Sports Activities:

- Secured **third** place in the **District-Level Karate Competition** held in Madhya Pradesh in 2019.